WORKED SOLUTION

STEP II 2016, Question 3

Pure • Roots of Polynomials

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q3

The key idea

Everything rests on one identity: $\mathrm{f}_n(x) = \mathrm{f}_{n-1}(x) + \dfrac{x^n}{n!}$. Combined with part (i), it says that at a root a of f_n the slope is $\mathrm{f}_n'(a) = -a^n/n!$, so we know the gradient at every root without knowing the roots. Since all roots have the same sign, all those gradients have the same sign, and a curve cannot cross the axis upwards twice in a row without crossing downwards in between.

Solution

Throughout, write $\mathrm{f}_n(x) = \displaystyle\sum_{k=0}^{n} \frac{x^k}{k!}$.

Part (i)

Differentiating term by term, the constant term 1 disappears and, for $1 \le k \le n$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^k}{k!}\right) = \frac{kx^{k-1}}{k!} = \frac{x^{k-1}}{(k-1)!}.$$

So

$$\mathrm{f}_n'(x) = \sum_{k=1}^{n} \frac{x^{k-1}}{(k-1)!} = \sum_{j=0}^{n-1} \frac{x^j}{j!} = \mathrm{f}_{n-1}(x),$$

as required.

Part (ii)

If $x \ge 0$, every term of $\mathrm{f}_n(x)$ is non-negative and the first term is 1, so $\mathrm{f}_n(x) \ge 1 > 0$. Hence no root can satisfy $a \ge 0$, and any real root has $a < 0$. (For $n = 0$, $\mathrm{f}_0 = 1$ has no roots at all, which is consistent.)

Part (iii)

From the definition, $\mathrm{f}_n(x) = \mathrm{f}_{n-1}(x) + \dfrac{x^n}{n!}$. If $\mathrm{f}_n(a) = 0$, then by part (i)

$$\mathrm{f}_n'(a) = \mathrm{f}_{n-1}(a) = \mathrm{f}_n(a) - \frac{a^n}{n!} = -\frac{a^n}{n!},$$

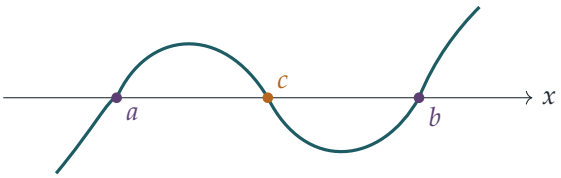
and likewise $\mathrm{f}_n'(b) = -b^n/n!$. Therefore

$$\mathrm{f}_n'(a)\,\mathrm{f}_n'(b) = \frac{(ab)^n}{(n!)^2} > 0,$$

because $a < 0$ and $b < 0$ by part (ii), so $ab > 0$. This is the required inequality. In particular neither gradient is zero, and the two have the same sign.

Ask yourself Why does equal sign of the two gradients force another root in between?

Take $a < b$ and suppose both gradients are positive (the negative case is the mirror image). Since $\mathrm{f}_n'(a) > 0$, the curve crosses the axis upwards at a, so $\mathrm{f}_n(x) > 0$ for x slightly bigger than a. Since $\mathrm{f}_n'(b) > 0$, it also crosses upwards at b, so $\mathrm{f}_n(x) < 0$ for x slightly smaller than b. A continuous function that is positive at some point and negative at a later point must vanish in between (intermediate value theorem), so $\mathrm{f}_n(c) = 0$ for some c with $a < c < b$.



The curve rises through a and rises through b, so it must come down through the axis somewhere between them.

At most one real root. Suppose $(*)$ had two or more distinct real roots. It is a polynomial of degree n, so it has finitely many roots, and we may choose two of them, $a < b$, that are *consecutive*, with no root strictly between them. But the argument above produces a root c strictly between a and b, a contradiction. So $(*)$ has at most one real root. (The argument used $n \geq 2$ as in the question; for $n = 1$ the only root is -1, and for $n = 0$ there is none, so the conclusion holds for every n.)

n odd. Now f_n has odd degree with positive leading coefficient $1/n!$, so $\mathrm{f}_n(x) \to -\infty$ as $x \to -\infty$ and $\mathrm{f}_n(x) \to +\infty$ as $x \to +\infty$. By the intermediate value theorem it has at least one real root, and so **exactly one**.

n even. For $n = 0$ there are no roots. For even $n \geq 2$, $\mathrm{f}_n(x) \to +\infty$ as $x \to \pm\infty$, so f_n attains a global minimum at some point x_0, where $\mathrm{f}_n'(x_0) = \mathrm{f}_{n-1}(x_0) = 0$. Since $\mathrm{f}_{n-1}(0) = 1 \neq 0$, $x_0 \neq 0$, and so

$$\mathrm{f}_n(x_0) = \mathrm{f}_{n-1}(x_0) + \frac{x_0^n}{n!} = \frac{x_0^n}{n!} > 0,$$

because n is even. The minimum value is positive, so $\mathrm{f}_n(x) > 0$ for all x and $(*)$ has **no real roots**.

(Alternatively: a root a would be simple, since $\mathrm{f}_n'(a) \neq 0$, so f_n would change sign there; but an even-degree polynomial with positive leading coefficient that changes sign exactly once would have to be negative for all large x or all very negative x, which is impossible.)

Checks

- $n = 1$: $1 + x = 0$ has the single root -1. $n = 2$: $1 + x + \frac{1}{2}x^2 = \frac{1}{2}\big((x+1)^2 + 1\big) > 0$, no roots. Both fit.
- $n = 3$: the root is about -1.596, and there $\mathrm{f}_3'(a) = \mathrm{f}_2(a) \approx 0.678 = -a^3/3!$, as predicted.
- A computer check of the sign changes of f_n for every $n \leq 25$ found exactly one root for odd n and none for even n.

Common slips

- Showing the product of gradients is positive without saying *why* $ab > 0$: it needs part (ii).
- Concluding "at most one root" from a root between any two roots without choosing the two roots to be consecutive; that choice is what produces the contradiction.
- For even n, forgetting to check $x_0 \neq 0$, which is needed for $x_0^n/n! > 0$ to be strict.

Beyond STEP

The polynomials f_n are the Taylor polynomials of e^x, and part (i) is the finite shadow of $(\mathrm{e}^x)' = \mathrm{e}^x$. Since e^x never vanishes, it is natural that the f_n have few real roots; their complex roots, scaled by $1/n$, cluster on a curve called the Szegő curve as $n \to \infty$. The tools used here, the intermediate value theorem and the existence of a minimum for a continuous function, are proved carefully in a first analysis course; see Abbott, *Understanding Analysis*.

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